

Alloys

Learning outcomes

By the end of this section you should be able to:

- Define the term alloy.
- Compare how the structure and properties of an alloy differs from its base metal.
- Explain the properties of alloys in terms of non-directional bonding.

When you look around your kitchen, your household appliances, city skyscrapers, car bodies or tools, can you identify what metal many of these things are made from? Many metallic objects commonly encountered are made from steel as shown in **Interactive 1**. Scroll through the photographs to see what they are.



 Rights of use

Interactive 1. Examples of Commonly Encountered Materials Made of Steel.

 More information for interactive 1

Steel is one of the most commonly used materials in the world. Have you heard of steel before? What about stainless steel? If you look on the periodic table, do you see steel listed as an element? What does this suggest that steel is? Do you know what elements or combination of elements make steel? What is it about the metallic structure and bonding of steel that makes this metal so useful?

Recall from [section S2.3.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-bonding-and-structure-id-43464/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-bonding-and-structure-id-43464/>) that the metallic bond is the electrostatic attraction between cations in a lattice and delocalised valence electrons. When we looked at ionic and covalent bonding, we studied the bonding between atoms of different elements. In [section S1.1.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>) you learned that atoms of different elements bonding together in a fixed ratio forms a compound. But when we looked at metals, we only considered the bonding between atoms of the same element. Why do you think this is?

When metals combine with other metals, they are less likely to form compounds, instead they form something called an alloy. Alloys are mixtures of a metal and other metals or non-metals. In general, alloys are made up of a base metal with additional elements added. This is illustrated in **Figure 1**. The addition of another element results in a metal with different properties than the base metal. Steel is an example of an alloy that is a mixture of iron with small amounts of carbon. Note that we can also use the triangle of bonding we looked at in [section S2.4.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bonding-triangle-id-43843/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bonding-triangle-id-43843/>) to analyse the type of bonding for alloys made up of only two elements.

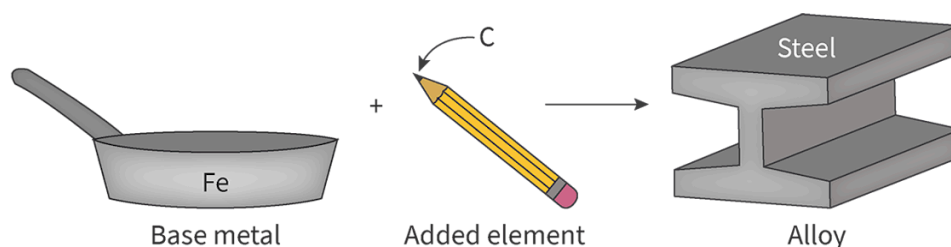


Figure 1. An alloy is made up of a base metal with small amounts of additional elements.

 More information for figure 1

The image is an illustration depicting the formation of alloys. On the left, there's a frying pan with the label 'Fe', representing iron. In the middle, there's a yellow pencil with a pink eraser, typically made of other elements. On the right, there is a structural steel beam labeled 'Steel', illustrating a mixture of iron and carbon. This diagram highlights the transition from a pure base metal (Fe) to an alloy (Steel) with altered properties due to the addition of other elements.

[Generated by AI]

Making connections

In [section S1.1.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>) you learned about compounds and mixtures. Would you expect alloys to be an example of homogeneous or heterogeneous mixtures?

Alloys are homogeneous mixtures as they have a uniform appearance and composition throughout.

Note that in this section we have referred to alloys as mixtures instead of as compounds. We can distinguish between the two terms:

- Compound — atoms of different elements bonded together in a fixed ratio.
- Mixture — composed of two or more substances in no fixed ratio.

Why are alloys more correctly described as mixtures rather than as compounds?

Structure of alloys

Recall from section S2.3.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-bonding-and-structure-id-43464/>) that the structure of elemental metals is a lattice of positive metal ions surrounded by delocalised valence electrons. The structure of alloys is similar to elemental metals, the key difference being that the added element will vary in size from the metal making up the base metal. There are two main types of alloys that form:

- Substitutional alloy - where the element added to the base metal simply replaces the metal ions in the lattice.
- Interstitial alloy - where the element added occupies vacant space in the metallic lattice of the base metal.

In both of these types of alloys, bonding is non-directional since the electrostatic force between cations and delocalised electrons still occurs in all directions. These two types of alloys are compared with that of an elemental metal below in **Figure 2**.

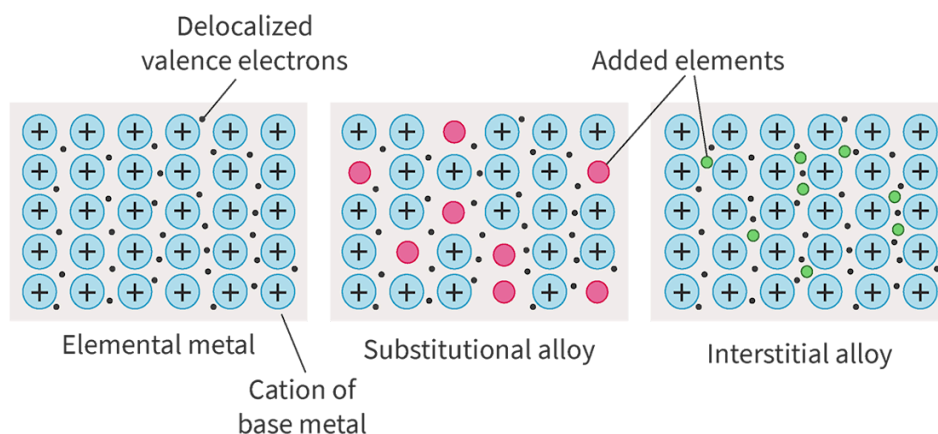


Figure 2. Comparison of the structure of an elemental metal, a substitutional alloy and an interstitial alloy.

 More information for figure 2

The image is a diagram illustrating the comparison between the structures of an elemental metal, a substitutional alloy, and an interstitial alloy. On the left, the elemental metal structure is shown with blue circles representing metal cations and gray dots depicting delocalized electrons. In the center, the substitutional alloy is depicted with

blue circles as the primary metal and red circles indicating substitute metal atoms, illustrating how different metal atoms can replace each other in the lattice. On the right, the interstitial alloy is illustrated with smaller green circles situated within the gaps of the metal lattice, indicating how smaller atoms occupy interstitial spaces. Labels describe each type of structure above the corresponding illustration.

[Generated by AI]

Study skills

You do need to describe alloys as mixtures of metals with other metals or non-metals and explain the properties of alloys in terms of non-directional bonding. The type of alloy (substitutional and interstitial) can be helpful in aiding understanding, but you are not required to recall these types in your examination.

In the structures of alloys in **Figure 2**, you can see how the base metal and the added element are not structurally arranged in a fixed ratio. Furthermore, since the bonding is between the ions in the lattice and the delocalised electrons in all directions, bonding is not between elements. This means there is no fixed ratio of elements bonded together. This is why alloys are more appropriately described as mixtures rather than compounds.

Physical properties of alloys

As you might expect, the introduction of different elements into the lattice of the base metal will influence the physical properties of the resultant material. Two physical properties that are most commonly affected through the process of alloying are:

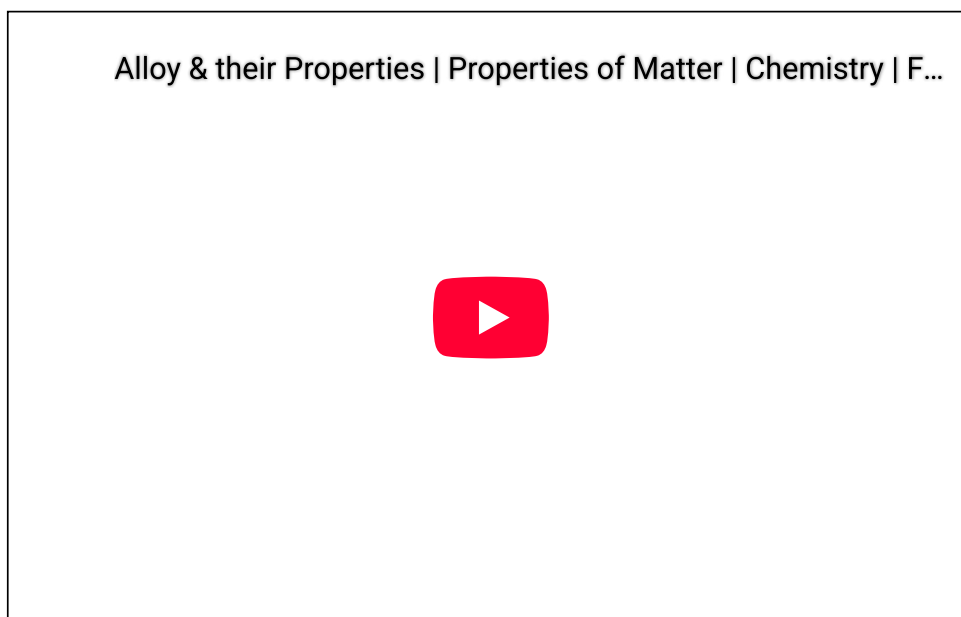
- Malleability/hardness
- Melting point.

Let's look at each of these individually.

Malleability/hardness

In section S2.3.1b (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-properties-and-application-of-metals-id-43563/>) we looked at how metals were malleable because layers in the lattice could easily slide past one another in response to an applied stress due to the non-directional bonding. What do you think will happen to the malleability of a metal when additional elements are added into the lattice structure to create an alloy? Would you expect layers to be able to slide past one another more easily, making the metal more malleable, or would it be more difficult for layers to slide past one another, making the alloy harder than the base metal?

The addition of elements with different sizes prevents the layers in the metallic lattice from sliding past one another easily. This causes the alloy to be less malleable and harder than the base metal. This is illustrated in **Video 1**.



Video 1. How alloys alter the properties of metals.

Melting point

Recall from [section S2.3.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-the-metallic-bond-id-43564/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-the-metallic-bond-id-43564/) that the stronger the electrostatic attraction between metal cations and delocalised electrons, the stronger the metallic bond and the higher the melting point. How might you expect the melting point of an alloy to differ from that of its base metal? How might the introduction of other cations into the metallic lattice influence the strength of the electrostatic attraction between cations and delocalised electrons?

Table 1 shows a comparison of the melting point temperatures for several elemental metals and alloys with those base metals.

Table 1. Comparison of the melting point temperatures between elemental metals and all

Elemental metal		Alloy	
Metal	Melting point (°C)	Alloy	Approximate melting point
Copper	1085	Bronze (copper with tin)	913
		Brass (copper with zinc)	930

Elemental metal		Alloy	
Metal	Melting point (°C)	Alloy	Approximate melting point
Iron	1538	Steel (iron with carbon)	1515
		Stainless steel (iron with chromium, and nickel)	1425
Aluminium	660.3	Aluminium alloy (aluminium with copper, manganese, nickel, zinc)	565
Tin	231.9	Solder (50% tin, 50% lead)	215
Lead	327.5		

Look at the melting points of tin and lead in **Table 1**. Now look at the approximate melting point for solder, an alloy made of tin and lead. What do you notice about the melting point for these two metals when they are combined as an alloy?

Notice that it is lower. What about the other alloys in **Table 1**?

The addition of elements into the lattice structure of a base metal will influence the strength of the metallic bond. For many alloys, this lattice structure with varying cation size weakens the electrostatic attraction between the cations and the delocalised electrons. This causes many alloys to have lower melting temperatures than the base metal due to this weaker metallic bond.

You may also have noticed in **Table 1** that the melting point for alloys is listed as an approximate temperature. This is because the melting point will change depending on the amount of alloying element added. Chemists studying alloys regularly make use of a phase diagram which helps to track the melting point of alloys as the percent composition is altered. To construct a phase diagram, scientists would study what temperature the alloy of each percentage composition would:

1. show signs of melting
2. complete melting.

Figure 3 is an example of a phase diagram for a tin-lead alloy, commonly known as solder.

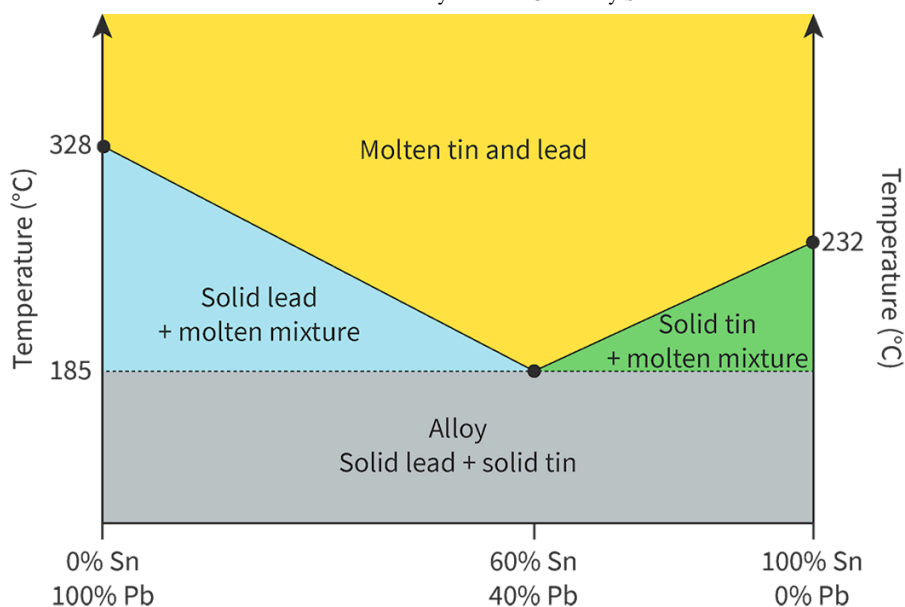


Figure 3. Phase diagram for tin-lead alloys.

🔗 More information for figure 3

Practical skills

Tools 1: Experimental techniques — Addressing safety

Inquiry 1: Exploring and designing — Exploring

Tin and lead are both examples of metals with low melting points. This means that you can melt these metals to create alloys easily using hot plates commonly found in a high school chemistry laboratory. Can you think of ways you could safely design an experiment to create a tin-lead alloy of varying percent composition? What equipment would you need? How might you study the melting temperature of your alloy either quantitatively or qualitatively? What are the safety considerations you would need to take to handle a substance like lead that has known toxicity concerns?

Common examples of alloys

Many other physical properties of alloys vary from their base metals, but they tend to be unique to specific alloys as opposed to all alloys. Let's look at some common examples of alloys and their unique physical properties in **Table 2**.

Table 2. Common examples of alloys and their unique physical properties.

Alloy	Constituent elements	Physical properties	Uses
Steel	Fe + C	High strength, very hard material	Constructive materials
Stainless steel	Steel + Cr, Ni	Resistant to corrosion Metal does not rust as easily	Kitchenware surfaces that require to be cleaned regularly
Bronze	Cu + Sn	Hard material, low melting point	Weapons and armour
Brass	Cu + Zn	Resistant to corrosion	Musical instruments, gears, hinges, valves
Aluminium alloy	Al + Cu, Mn, Ni, Zn	High strength, resistant to corrosion Low density (due to low density of Al)	Aerospace, aircraft, gas cylinders, scuba diving tanks
Solder	Sn + Pb	Low melting point More conductive than Pb but less conductive than Sn	Connecting electrical wires, plumbing pipes
Amalgam	Hg + Ag, Sn, Cu	Hardens quickly upon formation of alloy Resistant to corrosion	Dental fillings
Sterling silver	Ag + Cu	Strong and hard Resistant to tarnish	Victorian era cutlery, sets and jewellery
White gold	Au + Pd, Ni, Zn	Hard and strong Light grey colour	Jewellery
Rose gold	Au + Cu	Hard and strong. Red hue in colour	Jewellery

You may have noticed that many of the alloys in **Table 2** have resistance to corrosion as a physical property. Corrosion is the deterioration of a material on its surface as it interacts with substances in the surroundings. For metals, corrosion is typically the formation of brittle metal oxides on the surface due to the process of oxidation. You will learn more about the process of oxidation in [section R3.2.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/).

Study skills

Note that you do need to be able to explain why the properties of alloys differ from their base metals. However, you are not required to know specific examples of alloys. These examples simply aid in the understanding of these differing properties.

In this next activity you will research and become an ‘expert’ on one specific alloy that you will teach your group all about.

Activity

- **IB learner profile attribute:** Risk-taker
- **Approaches to learning:** Research skills — Using search engines and libraries effectively
- **Time required to complete activity:** 45—60 minutes
- **Activity type:** Individual activity for #1, groups of 4—5 for #2

Alloy research jigsaw activity

1. Home groups

In groups of 4—5, look through the following list of alloys. Each group member will select one alloy to research and become an ‘expert’ on.

	Types of Alloys
Alloys	• Hyduminium (aluminium based alloy) • Wood’s metal (bismuth based alloy) • Nichrome (nickel based alloy) • Pewter (tin based alloy) Galinstan (gallium based alloy)

2. Expert groups

Create a new expert group with individuals in the class that have chosen the same alloy as you. In this expert group you will research the specified alloy as a team (approximately 10 minutes). Research the following:

- (a) Composition of alloy.
- (b) The approximate melting point of the alloy.
- (c) Unique properties this alloy has in comparison to its base metal.

(d) Examples of where this alloy is used.

Sketch a diagram representing the structure of the alloy you have researched.

As expert groups, evaluate information sources you are using for accuracy, bias, credibility and relevance. You will be sharing your research findings with your home group. You may want to take notes during your research to help you with this.

3. Return to home groups

Returning to your home group, each member of the group will share research findings of their alloy. You may consider using a slideshow, chart paper, whiteboard, etc. to share your research findings.